Study Notes

T, F and Chi Square tests

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1. T-Test

Purpose:
In econometrics, the T-test is used to test hypotheses about individual coefficients in a regression model, particularly to assess whether a specific coefficient differs significantly from zero, which would indicate that the associated independent variable has a significant impact on the dependent variable.

Application:

- **Testing Individual Regression Coefficients**: The T-test is used to test the null hypothesis that a particular coefficient $\beta i=0$ against the alternative hypothesis $\beta i\neq 0$. This is done by calculating the t-statistic and comparing it against critical values from the t-distribution.

Formula:

- $t=b/SE(b)$, where b is an estimator of βi.

Interpretation:

- If the calculated t-value is greater than the critical t-value from the t-distribution table at a chosen significance level (e.g., 0.05), the null hypothesis is rejected, indicating that the coefficient is statistically significant.

2. F-Test

Purpose:

The F-test in econometrics assesses the overall significance of a regression model. It tests whether at least one of the coefficients in the regression model is non-zero, implying that the independent variables, collectively, explain a significant portion of the variance in the dependent variable.

Application:

- **Testing Overall Model Significance**: The F-test is used to test the null hypothesis that all regression coefficients (except the intercept) are equal to zero, meaning that none of the independent variables have a linear relationship with the dependent variable.

Formula:

- $F=(ESS/(k-1))/(RSS/(n-k))$

Where:

- ESS= Explained Sum of Squares

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- RSS = Residual Sum of Squares
- k = Number of parameters (including intercept)
- n = Number of observations

Interpretation:

- If the calculated F-value is greater than the critical value from the F-distribution table, the null hypothesis is rejected, indicating that the model is statistically significant.

3. Chi-Square Test

Purpose:

The Chi-Square Test is used to assess the association between categorical variables. In econometrics and statistics, it determines whether there is a significant association between the observed frequencies in certain categories and the expected frequencies, assuming no association (null hypothesis).

Application:

- Test of Independence: Used to determine if two categorical variables are independent.
- Goodness-of-Fit Test: Used to test if an observed frequency distribution fits a specified distribution.

Formula:

$$\chi^2 = \sum (O_i - E_i)^2 / E_i$$

Where:

- Oi = Observed frequency for category i
- Ei = Expected frequency for category i

Interpretation:

- Degrees of Freedom (df): Calculated as (r−1)×(c−1) for a test of independence where r and c are the number of rows and columns.
- If the calculated Chi-Square statistic exceeds the critical value from the Chi-Square distribution table at a chosen significance level (e.g., 0.05), the null hypothesis is rejected, indicating a significant association between the variables.

This test is typically used in contexts like survey analysis, quality control, and other areas where categorical data is analyzed.

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